Transfer Learning from Fermi to Swift for GRB Redshift Prediction

A feasibility note: what we are transferring, how it is done, and what we found
Prepared for the Dainotti GRB collaboration · October 2026

Bottom line. We tested whether prompt gamma-ray emission information from Fermi-GBM — in particular the spectral peak energy E_{peak} — can improve the Swift X-ray redshift estimator, by "transferring" it from the Fermi domain into the Swift domain using the 67 bursts observed by both instruments. We tried three ways of doing the transfer. **None improves redshift prediction.** The reason is physical, not a coding choice: the *observed* E_{peak} is essentially uncorrelated with redshift ($r = 0.02$ across 163 bursts), so there is no redshift signal to transfer in the first place.

1. What are we trying to do?

The existing estimator predicts a gamma-ray burst's redshift from Swift X-ray "plateau" features — properties of the slowly-declining X-ray afterglow (the plateau flux and end-time, the decay slope, the absorbing column, and so on). It reaches a correlation of about $r = 0.53$ between predicted and true redshift on the 281-burst sample.

The proposal was to give the model *new* information from a different phase of the burst: the **prompt** gamma-ray emission, which Fermi's Gamma-ray Burst Monitor (GBM) records. In particular, the Fermi prompt spectrum has a well-defined **peak energy, E_{peak}** — the photon energy at which the burst radiates most of its power. E_{peak} is physically central to GRBs (it appears in the famous Amati and Yonetoku relations), so it is a natural thing to try to fold in.

The obstacle: almost none of the 281 Swift GRBs have Fermi measurements. We cannot simply paste an E_{peak} column onto the Swift table. We need a way to carry E_{peak} from the instrument that measures it (Fermi) into the sample that does not (Swift). **That bridge is what "transfer learning" refers to here.**

2. What is "transfer learning" in this context? (question a)

In general, **transfer learning** means reusing knowledge gained in one setting — the *source domain* — to help a task in a different but related setting — the *target domain*. It is most useful when the target has limited data, or is missing a measurement that the source has. In our case:

- **Source domain:** Fermi-GBM, the prompt gamma-ray phase. It measures E_{peak} (and other prompt spectral quantities).
- **Target domain:** Swift, the X-ray afterglow phase. It has the 281-burst redshift sample, but no E_{peak}.

- **The bridge:** 67 GRBs observed by *both* instruments. For these we have the Swift features, the Fermi E_{peak}, and a measured redshift all at once. They are what lets us learn how the two domains relate.

So, to answer the question directly: **what we are transfer-learning is E_{peak}** (and, more broadly, prompt-emission information). We are trying to take a quantity that lives in the Fermi world and make it usable in the Swift world, using the 67 common bursts to learn the connection. This mirrors the group's earlier *optical → X-ray* transfer, which brought optical-plateau information into the X-ray analysis; here the source is prompt emission instead of optical.

3. The data

From Maria's spreadsheet (Sheet 4), 67 GRBs are common to the Swift X-ray plateau sample and the Fermi-GBM catalog *and* have a measured redshift. These are the bridge bursts. For each one we retrieved the Fermi prompt spectral fit from the official Fermi-GBM Burst Catalog (the HEASARC `fermigbrst` table), using the time-integrated Comptonized model (FLNC COMP), as specified: E_{peak}, the prompt photon index, the prompt fluence and flux, each with its uncertainty.

Of the 67, **60 have a well-constrained E_{peak}**; the rest have an E_{peak} that the fit cannot pin down (their spectra are too featureless within the GBM band, so the "peak" runs off to implausible energies). We trained on the 60 good ones. For reference, a reminder of what E_{peak} is: the energy at which the prompt spectrum peaks in nu-F-nu; the observed values here span roughly 9 to 2000 keV.

4. How is the transfer done? (question b)

We tried three versions, from the simplest sanity check to the full "combine the samples" idea.

4a. Direct check (on the bridge bursts only)

The simplest test. On the 60 bridge bursts — where we have both Swift features *and* a measured E_{peak} — we just add E_{peak} as an extra column to the redshift model and ask whether prediction improves. This is not something we could deploy (it only works where E_{peak} is actually measured), but it is the essential prerequisite: *if measured E_{peak} does not help even here, nothing fancier will.*

4b. Transfer map / imputation (the main version)

This is the literal "obtain E_{peak} via transfer learning" idea. In three steps: (1) on the 60 bridge bursts, learn a model that predicts E_{peak} from the Swift observables; (2) apply that model to all 281 Swift bursts to *impute* an E_{peak} for each, even though they have no Fermi data; (3) add the imputed E_{peak} to the redshift model and check whether prediction improves.

4c. Pooled / domain adaptation (combine the samples)

Rather than imputing one variable, this combines the two *samples* into one larger training set. Because the two instruments do not measure the same things, we can only pool on the variables they *share* — the prompt photon index, the duration T90, the fluence, and the peak flux — plus a 0/1 “instrument” indicator (a *dummy variable*) so the model can absorb systematic differences between Swift and Fermi. The hope is that extra training data, from a related instrument, sharpens the predictions.

5. What we found

All three approaches give essentially no improvement. The clearest single comparison is the real estimator (the winning GLM, with its squared and interaction terms, on the full feature set) with versus without the transfer-learned E_{peak}:

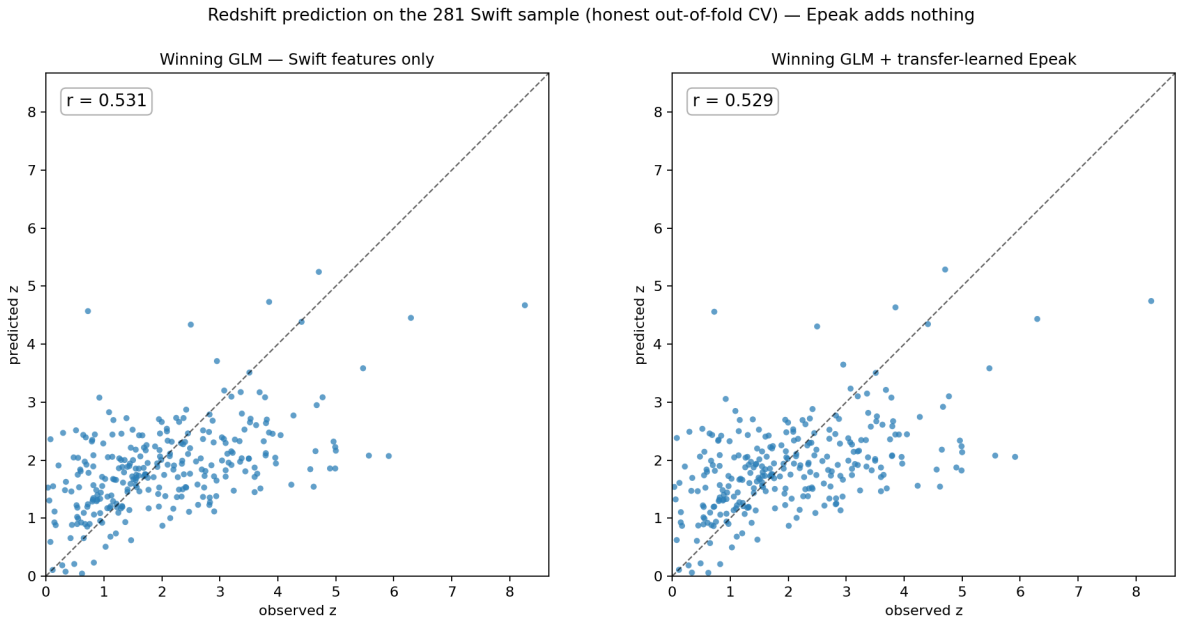


Figure 1. Predicted vs. true redshift on the 281 Swift bursts (honest cross-validation). Left: the estimator as it stands. Right: the same estimator after adding the transfer-learned E_{peak}. The two are point-for-point indistinguishable (r = 0.531 vs. 0.529).

To avoid a confusion that is easy to fall into, here is the full ladder of numbers, with each row labelled by *which model* and *which features* it uses. The headline 0.53 is the winning GLM on the full feature set; the lower numbers further down are not failures of the transfer — they come from deliberately using a simpler model or a smaller feature set in a particular test.

Setup (281-burst sample, honest CV)	r
Winning GLM (squares + interactions), full plateau features ← the headline	0.53
Winning GLM + transfer-learned Epeak (test 4b)	0.52
Plain linear model, full 7 plateau features	0.46

Plain linear model, only the 4 shared variables (basis for test 4c)	0.29

Table 1. The two steps down from 0.53 to 0.29 are (i) using a plain linear model instead of the GLM (−0.07, the GLM's nonlinear terms carry real signal) and (ii) restricting to the four variables Fermi and Swift share, which discards the strong plateau features (−0.17). The pooled test (4c) necessarily lives in that bottom, shared-variable world.

The precise results of each approach: **4a** (measured E_{peak} on the 60 bridge bursts): change in r of about −0.04 to 0, buried in noise. **4b** (transfer-learned E_{peak}, winning GLM on 281): 0.518 → 0.515, and a nonlinear random-forest check gives the same null. **4c** (pooled, shared variables + dummy): 0.287 → 0.275 — pooling slightly *hurts*.

6. Why it does not help (the important part)

Three reasons, each worth understanding because they are the real conclusions:

Reason 1 — there is no redshift signal in observed E_{peak} to begin with.

We checked this directly on a much larger, cleaner sample: 163 Fermi GRBs with measured redshift. The observed E_{peak} is **uncorrelated with redshift** (Pearson r = 0.02, p = 0.8).

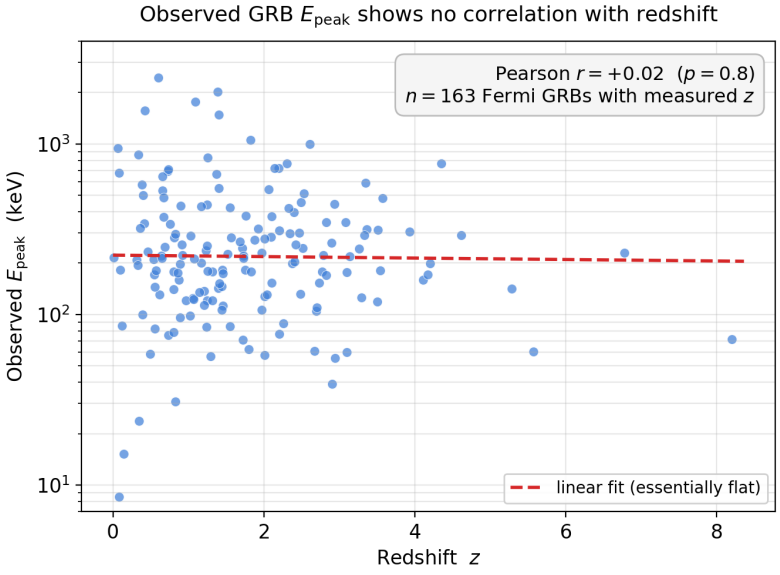


Figure 2. Observed E_{peak} vs. redshift for 163 Fermi GRBs. The relationship is flat.

The physics behind this is clean. The physically meaningful quantity is the *rest-frame* peak energy, $E_{peak,rest} = E_{peak,obs} \times (1+z)$. The Amati and Yonetoku relations link $E_{peak,rest}$ to a burst's energy or luminosity — but using them requires already knowing z. In the *observer* frame, the large intrinsic spread in E_{peak} from burst to burst swamps the (1+z) stretching, so the observed E_{peak} tells you almost nothing about redshift. This one fact explains every null result below it: **you cannot transfer information that is not there.**

Reason 2 — the imputed E_{peak} is redundant with the Swift features (a circularity).

In the transfer-map approach (4b), the imputed E_{peak} is *computed from* the Swift features. Feeding a model a new column that is just a function of columns it already has gives it no new information. We confirmed this concretely: the imputed E_{peak} can be reconstructed from the Swift features with $r = 1.0$, and adding it changes the GLM by -0.004 and the random forest by -0.003. Importantly, this holds for *any* model — linear, GLM, or a nonlinear ensemble like the SuperLearner — because it is about information content, not the model's flexibility.

Reason 3 — the pooled approach can only use the weak shared variables.

To pool with Fermi you may only use variables both instruments measure. But Swift's predictive power lives in the afterglow *plateau* features (the plateau flux and time), which Fermi does not have. Dropping to the shared prompt-type variables alone costs you most of the signal ($0.46 \rightarrow 0.29$) before any transfer even happens. Worse, the shared variables are not quite the same physical quantities across instruments (Swift's photon index is from the X-ray *afterglow*; Fermi's is from the *prompt* gamma rays), so their relationships to redshift differ and the pooled data pulls the fit slightly off.

7. Caveats, in fairness

- **Small bridge sample.** The direct test (4a) uses only 60 bursts and is genuinely underpowered — which is exactly why we leaned on the 163-burst correlation (Figure 2), which is well-powered and model-free.
- **Read the p-values loosely.** The significance values from repeated cross-validation are anti-conservative (the resamples reuse the same bursts), so they indicate that an effect's *direction* is stable, not that it is strongly significant. The effect *sizes* are tiny regardless.
- **Model choice.** We used the time-integrated Comptonized E_{peak}, as specified. The power-law same-model variant is a possible follow-up, but it would not change the central fact that observed E_{peak} is uncorrelated with z.
- **"No detectable benefit" is not "mathematically impossible."** But the observer-vs-rest-frame argument makes a benefit from observed E_{peak} specifically very unlikely.

8. What would change the answer

A real gain would require a prompt-emission quantity that (a) Swift does not already have and (b) correlates with redshift *in the observer frame*. Observed E_{peak} fails (b). The genuinely useful E_{peak} relations are rest-frame, and using them needs z — which is circular for a redshift predictor. The other prompt variables we looked at (peak flux, fluence) were also flat-to-null, though they could be examined more carefully. If a substantially larger common sample became available, the direct test could be repeated with more statistical power — but the flat 163-burst correlation already argues strongly

against a payoff.

Data and figures underlying this note: the 67-burst bridge set (common67_transfer.csv), the 163-burst E_{peak}–redshift set (epeak_z_163.csv), and the two figures above, are in the project's new_data folder.